Statement of purpose

PhD application · Computer vision for medical imaging · Tania Rahman

In my MSc thesis I trained a model to find signs of diabetic eye disease using only 400 labelled images. Contrastive pre-training lifted the score from 0.78 to 0.86, but the model failed on images from a second clinic. That gap is the question I want to spend my PhD on: how to make label-efficient models stay reliable across hospitals.

My path to this question was practical. During my BSc I learned that the hardest part of medical AI is not the model but the data: labels are expensive, and every hospital images patients differently. I started a small project on public retina data, then joined a remote research assistantship where I reproduced three published methods and recorded exactly where each one broke. That work became a preprint and a poster, and taught me to report negative results honestly.

Your group's work on robust self-supervised pre-training for small medical datasets is the best place I know to study this. I have read your 2025 paper closely. Its open question on scanner shift is the one my experiments point to, and I propose to extend it with a cross-site evaluation using pooled unlabelled images, as outlined in my attached proposal.

I bring tested skills in PyTorch, careful experiment design and open, reproducible code, and I learn quickly: I began German this year to prepare for life and work in your lab's city. During my PhD I aim to publish a robust pre-training method and a benchmark of cross-site failures, and later to work with clinicians so that the models are used responsibly.

I would be grateful for the chance to discuss how my work could contribute to your group.

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